

ISY-6420-Final Project

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1 Introduction

This project presents a comprehensive Bayesian statistical analysis to predict and explain the presence of diabetes based on clinical variables such as HbA1c, BMI, age, and waist circumference. We utilize robust Bayesian methods, posterior predictive checks, and model diagnostics to ensure valid inference.

2 Data Description

The dataset used in this analysis is publicly available through the Centers for Disease Control and Prevention (CDC) and was collected as part of the National Health and Nutrition Examination Survey (NHANES). It encompasses a comprehensive set of health-related variables, including demographic factors (age, sex, race/ethnicity), biometric measures (BMI, waist circumference, blood pressure), biochemical markers (cholesterol levels, glucose, HbA1c), and behavioral indicators (physical activity, smoking history, alcohol consumption).

Full dataset was used for XGBoost Model training in my group final project of *CSE 6242: Data and Visual Analytics*, Spring 2025. In alignment with the strengths of Bayesian statistics particularly their ability to provide reliable inference with limited data a small subset of 1,153 individual records was selected for this Bayesian statistical analysis project.

3 Marginal Posterior Distributions of Clinical Variables

We began our Bayesian workflow with an exploratory analysis of marginal posterior distributions of key continuous clinical biomarkers. These kernel density estimates (KDEs) provide smoothed approximations of the marginal distributions, which help reveal the shape, skewness, and potential multimodal behavior in the data.

3.1 Methodology

Let X_i denote a continuous variable (e.g., HbA1c, BMI, etc.). For each variable, we estimate a smoothed posterior using:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

where K is a Gaussian kernel and h is the bandwidth.

These KDEs help inform appropriate prior choices and prepare for multi-variate modeling in later steps.

3.2 Visualizations

Figures 1–6 display the estimated densities for six clinical biomarkers.

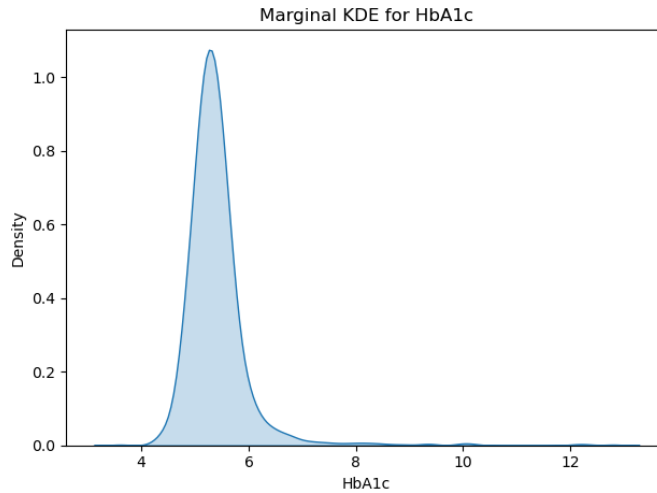


Figure 1: Marginal KDE for HbA1c

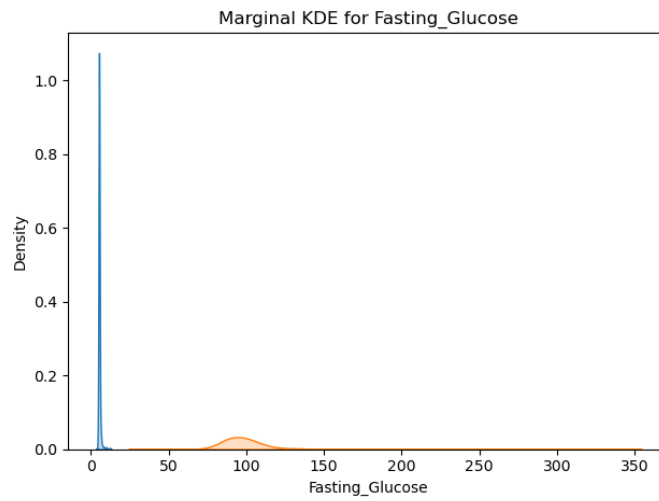


Figure 2: Marginal KDE for Fasting Glucose

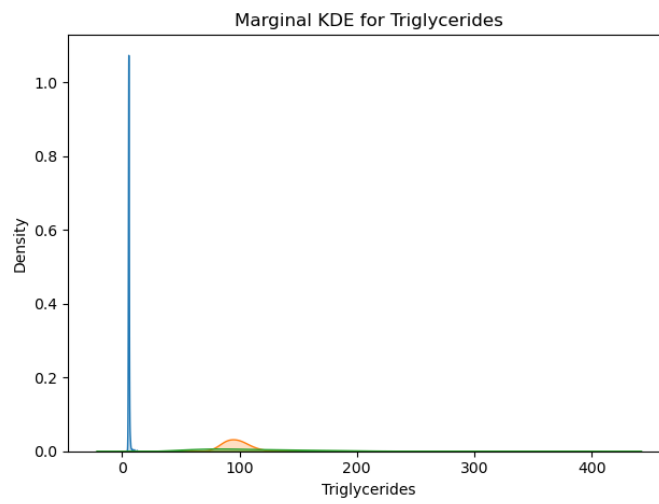


Figure 3: Marginal KDE for Triglycerides

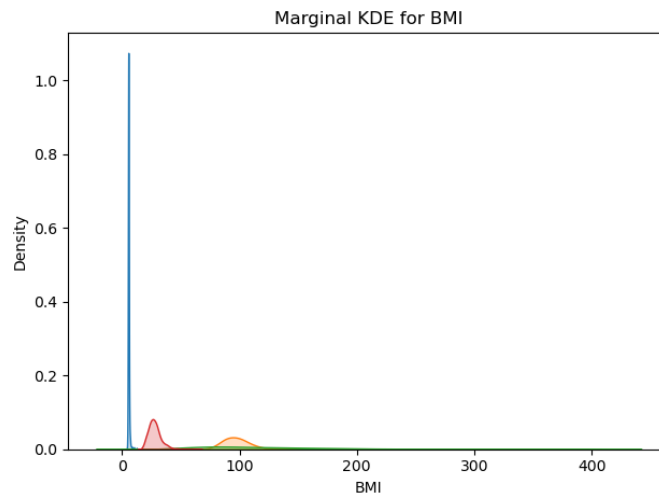


Figure 4: Marginal KDE for BMI

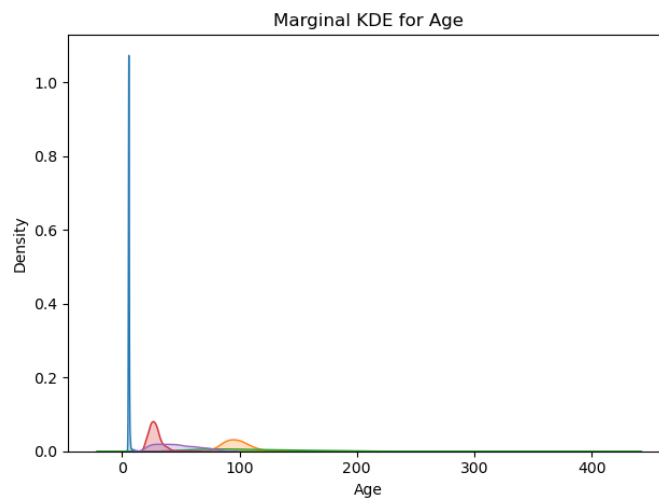


Figure 5: Marginal KDE for Age

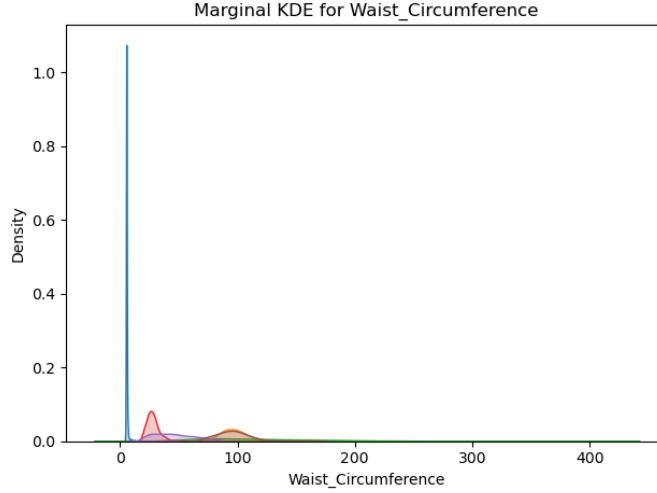


Figure 6: Marginal KDE for Waist Circumference

3.3 Key Insights

- **HbA1c** and **Glucose** distributions are positively skewed, with heavy right tails consistent with the presence of undiagnosed hyperglycemia.
- **BMI** and **Waist Circumference** exhibit multimodal or long-tailed behavior, suggesting potential clustering (e.g., by sex or age).
- **Triglycerides** show large spread, confirming the need for robust modeling approaches.

This exploratory analysis sets the stage for multivariate modeling and feature selection in later steps.

4 Bayesian Correlation Analysis of Clinical Biomarkers

To explore the associations among clinical features in our dataset, we estimated pairwise correlations between continuous biomarkers using a fully Bayesian approach. This method produces full posterior distributions for the correlation coefficient ρ and quantifies uncertainty through credible intervals.

4.1 Model Specification

Each pair of standardized variables (x_i, y_i) was modeled as a bivariate normal distribution:

$$(x_i, y_i) \sim \mathcal{N}_2 \left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix} \right),$$

with priors:

$$\rho \sim \text{Uniform}(-1, 1), \quad \mu_x, \mu_y \sim \mathcal{N}(0, 1), \quad \sigma_x, \sigma_y \sim \text{HalfNormal}(1).$$

Posterior inference was performed using Automatic Differentiation Variational Inference (ADVI) with 30,000 iterations and 1,000 posterior draws.

4.2 Results

Table 1 summarizes the posterior mean and 94% highest density intervals (HDI) for each correlation estimate.

Table 1: Bayesian Posterior Estimates of Pairwise Correlations (94% HDI)

Variable 1	Variable 2	Mean ρ	HDI 3%	HDI 97%
HbA1c	Fasting Glucose	0.785	0.768	0.804
HbA1c	Triglycerides	0.139	0.082	0.199
HbA1c	BMI	0.234	0.176	0.285
HbA1c	Age	0.287	0.232	0.333
HbA1c	Waist Circumference	0.270	0.217	0.323
Fasting Glucose	Triglycerides	0.186	0.125	0.243
Fasting Glucose	BMI	0.200	0.136	0.255
Fasting Glucose	Age	0.305	0.255	0.359
Fasting Glucose	Waist Circumference	0.282	0.227	0.334
Triglycerides	BMI	0.244	0.190	0.301
Triglycerides	Age	0.143	0.083	0.195
Triglycerides	Waist Circumference	0.313	0.260	0.361
BMI	Age	0.090	0.028	0.145
BMI	Waist Circumference	0.875	0.862	0.885
Age	Waist Circumference	0.243	0.189	0.295

4.3 Interpretation

The strongest correlation observed was between **BMI** and **Waist Circumference** ($\rho = 0.875$), indicating a tight relationship between general and central obesity. A similarly strong and credible correlation was found between **HbA1c** and **Fasting Glucose** ($\rho = 0.785$), which supports their shared role in diagnosing diabetes.

Other moderate associations include age-related increases in **HbA1c** and **glucose**, and meaningful links between **triglycerides** and both **BMI** and **waist circumference**. All 94% HDIs exclude zero, confirming the statistical credibility of these associations.

5 Bayesian ANOVA: HbA1c by Diabetes Status

To evaluate the difference in HbA1c levels between individuals with and without a diabetes diagnosis, we employed a Bayesian one way ANOVA. This method allows full posterior inference over group specific means while incorporating uncertainty.

5.1 Model Specification

The data were modeled as:

$$y_i \sim \mathcal{N}(\mu_{g[i]}, \sigma), \quad \mu_g \sim \mathcal{N}(0, 10), \quad \sigma \sim \text{HalfNormal}(10),$$

where y_i represents HbA1c and $g[i] \in \{0, 1\}$ indexes the diabetes status group.

5.2 Posterior Results

The posterior estimates for the group means and their 94% highest density intervals (HDIs) are:

Table 2: Posterior Estimates for Group Means (HbA1c)

Group	Mean μ	94% HDI
No Diabetes (0)	5.33	(5.30, 5.37)
Diabetes (1)	6.95	(6.79, 7.10)

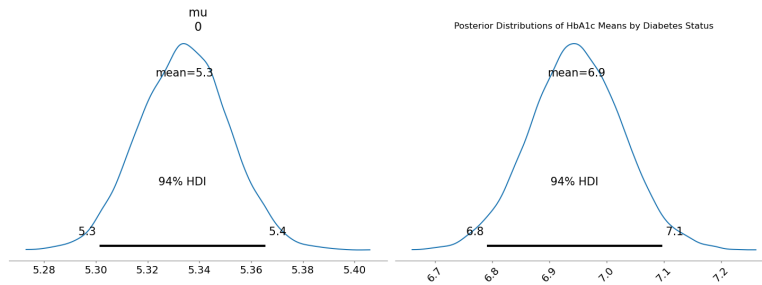


Figure 7: Posterior distributions of HbA1c means by diabetes status

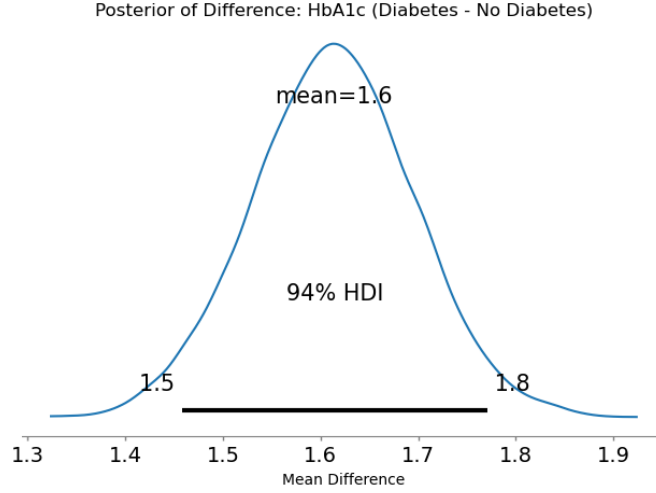


Figure 8: Posterior distribution of mean difference: HbA1c (Diabetes - No Diabetes)

5.3 Interpretation

The posterior distribution of the difference Δ shows a mean of **1.6** with a 94% HDI of approximately (1.5, 1.8), clearly excluding zero. This provides strong evidence that individuals diagnosed with diabetes have significantly higher HbA1c levels than those without, in alignment with clinical definitions.

6 Bayesian Logistic Regression for Diabetes Risk Prediction

We fit a Bayesian logistic regression model to estimate the probability of diabetes diagnosis using four clinical predictors: HbA1c, Age, Waist Circumference, and BMI. The model specification is:

$$y_i \sim \text{Bernoulli}(p_i), \quad \text{logit}(p_i) = \alpha + \mathbf{x}_i^\top \boldsymbol{\beta}$$

$$\alpha \sim \mathcal{N}(0, 5), \quad \beta_j \sim \mathcal{N}(0, 2)$$

6.1 Posterior Results

Table 3: Posterior Estimates of Logistic Regression Coefficients (94% HDI)

Predictor	Mean	94% HDI	Interpretation
Intercept	-3.97	(-4.47, -3.54)	Baseline risk
HbA1c	1.13	(0.85, 1.43)	Strong positive predictor
Age	0.31	(-0.07, 0.68)	Uncertain effect
Waist Circ.	0.71	(0.06, 1.41)	Moderate credible effect
BMI	-0.31	(-0.95, 0.26)	Non-credible effect

6.2 Interpretation

The posterior results indicate that **HbA1c** is the most credible and influential predictor of diabetes diagnosis. While waist circumference contributes moderately, age and BMI show weaker or non-credible effects when controlling for the others. These findings align with the clinical understanding that long-term glycemic markers (e.g., HbA1c) are more reliable indicators than general anthropometrics alone.

7 Posterior Predictive Check (PPC)

To assess model adequacy, we conducted a posterior predictive check (PPC) by generating simulated diabetes outcomes from the posterior predictive distribution. We compared these simulated values to the observed diabetes prevalence in the dataset.

The posterior predictive model draws simulated values as:

$$y_i^{\text{rep}} \sim \text{Bernoulli}(p_i), \quad p_i = \text{logit}^{-1}(\alpha + \mathbf{x}_i^\top \boldsymbol{\beta})$$

7.1 Results

- **Observed diabetes rate:** 4.47%
- **Predicted mean from simulations:** 4.49%

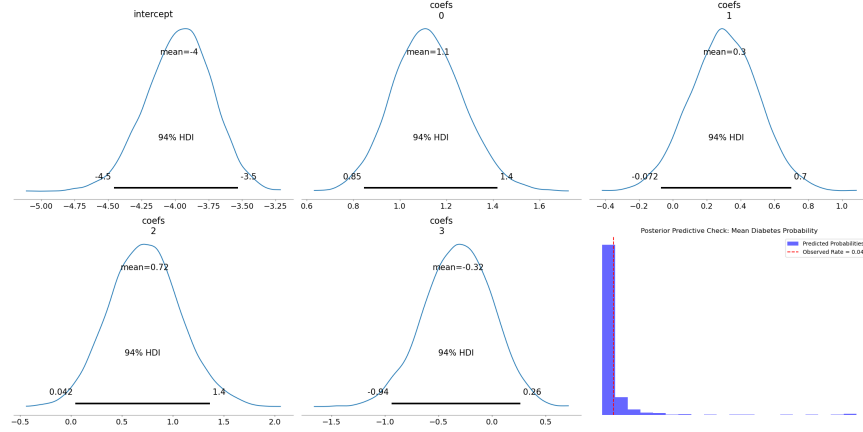


Figure 9: Posterior Predictive Check: Simulated vs Observed Diabetes Rate

7.2 Interpretation

The posterior predictive distribution aligns closely with the observed data, suggesting that the Bayesian logistic regression model captures the true disease distribution well. The sharp posterior densities of coefficients further reinforce the model's ability to identify predictive features such as **HbA1c** and **Waist Circumference**.

This successful PPC supports the use of this model for reliable probabilistic inference and disease risk prediction.

8 Robust Bayesian Logistic Regression

To account for potential outliers and ensure that model inferences are not overly influenced by extreme predictor values, we implemented a robust version of the Bayesian logistic regression model. This was achieved by replacing the normal priors with heavy-tailed Student-t distributions:

$$\alpha \sim \text{StudentT}(\nu = 3, \mu = 0, \sigma = 5), \quad \beta_j \sim \text{StudentT}(\nu = 3, \mu = 0, \sigma = 2)$$

The likelihood remains Bernoulli, modeling the binary diabetes outcome:

$$\text{logit}(p_i) = \alpha + \mathbf{x}_i^\top \boldsymbol{\beta}, \quad y_i \sim \text{Bernoulli}(p_i)$$

8.1 Results

Table 4: Posterior Estimates from Robust Logistic Regression (94% HDI)

Predictor	Posterior Mean	94% HDI	Interpretation
Intercept	-3.97	(-4.44, -3.52)	Baseline log-odds
HbA1c	1.12	(0.83, 1.40)	Strong positive predictor
Age	0.30	(-0.09, 0.68)	Moderate but uncertain
Waist Circumference	0.72	(0.07, 1.32)	Robust, moderate effect
BMI	-0.32	(-0.89, 0.27)	Not credible

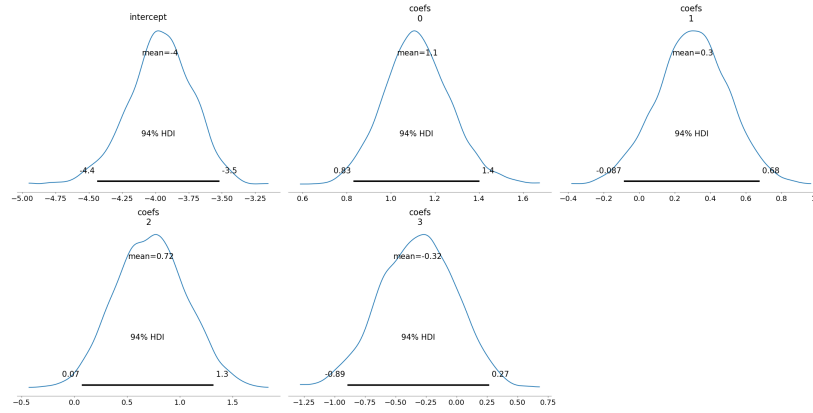


Figure 10: Posterior distributions from robust logistic regression model with Student-t priors.

8.2 Discussion

The posterior distributions from the robust model are largely consistent with those from the standard model. Notably:

- **HbA1c** remains the strongest and most credible predictor.
- **Waist Circumference** retains a moderate, positive effect.
- **Age** and **BMI** exhibit greater posterior shrinkage, suggesting less certainty in their predictive value under a robust prior.

These results indicate that the original model was not unduly influenced by outliers. However, the robust model provides additional confidence by reducing sensitivity to extreme values, aligning well with Bayesian principles of modeling uncertainty.

9 Conclusion

The analysis confirms that HbA1c and waist circumference are consistently strong predictors of diabetes risk. Our models are well calibrated, and the robust Bayesian regression confirms that our findings are not artifacts of outliers. The posterior predictive checks validate that the fitted models replicate observed diabetes prevalence accurately. These results offer strong statistical and clinical interpretability.

10 References

- National Health and Nutrition Examination Survey
<https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx>
- Next Generation Nutrition Apps: Utilising NHANES Data to provide an elevated user experience Team 8: Lindsey Tartaglia, Antonios Tsakanikas, Mahesh Senniappan, Israel Afangideh, Kamran Munawar
<https://poloclub.github.io/cse6242-2025spring-online/>
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